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EXTENDED ABSTRACT

GCMagicc: A lightweight climate model emulator with robust extrapolation across forcing scenarios and model distributions

Nicolai Meinshausen[1*], Malte Meinshausen[1], Xinwei Shen[1] and Elisabeth Vogel[1]

[1]Climate & Energy College, University of Melbourne, Melbourne, Australia
[2]Department of Statistics, UW Washington
*Corresponding author. Email: meinshausen@gmail.com

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1. Introduction and objectives

Climate model emulators approximate the output of general circulation models (GCMs) at a fraction of the computational cost. Most existing emulators either target global-mean quantities or fail to generalize to forcing scenarios outside the training distribution. We present GCMagicc, an emulator of gridded monthly climate fields with the following properties:

- Lightweight architecture that can be trained and evaluated efficiently.
- Preserves spatio-temporal structure across multiple variables without relying on externally specified temperature fields as input.
- Models the joint distribution of climate fields conditional on ten distinct forcing variables.
- Extrapolates to unseen model–scenario combinations without breaking down beyond the training support.
- Supports bias reduction relative to reanalysis data.
- Produces output at higher spatial resolution than the native GCM grid.

2. Methods and data

The emulator is trained on monthly output from approximately 30 CMIP6 models under historical, SSP1-2.6, SSP5-8.5, abrupt-4$\times CO_2$, and 1% CO_2 experiments, together with ERA5 reanalysis (Hersbach et al., 2020) for the historical period. Inputs consist of scalar effective radiative forcings (greenhouse gases, aerosols, ozone, solar, volcanic) and smoothed global-mean temperature and net radiative flux. Targets are ten gridded climate variables (surface pressure, 2 m temperature, precipitation, wind speed, surface temperature, daily max/min temperature, solar radiation, relative and absolute humidity) represented at multiple resolutions from 12 to 786 432 grid cells per field via a hierarchical decomposition. The generative model underlying GCMagicc relies on engression (Shen and Meinshausen, 2025), a distributional regression method with provable extrapolation guarantees, extended to high-dimensional outputs via reverse Markov learning (Shen et al., 2025). A related application of

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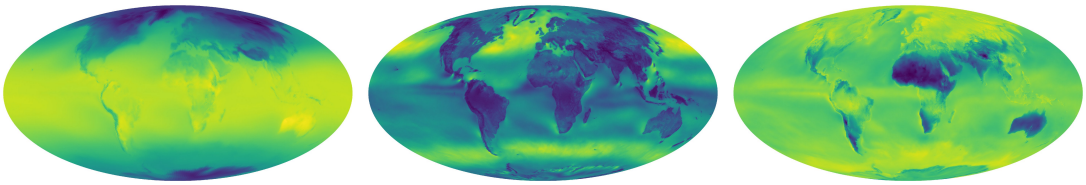

Figure 1. *Emulated fields generated by GCMagicc on a HEALPix grid with* $N_{\mathrm{side}} = 256$ *(*$786\,432$ *grid cells per variable). From left to right: 2 m temperature (K), wind speed (*$\mathrm{m\,s^{-1}}$*), and relative humidity (%)..*

these methods to statistical downscaling of regional climate model output is described in Schillinger et al. (2025).

3. Preliminary results

GCMagicc reproduces ERA5 spatial patterns under historical forcings for all ten variables (see Figure 1 for examples of emulated fields). Out-of-distribution evaluation on held-out models (e.g. UKESM under SSP5-8.5) shows good agreement in both global-mean trajectories and spatial quantiles. Intervention experiments, such as removing aerosol forcing under present-day conditions, yield physically plausible warming patterns concentrated over the Northern Hemisphere. The emulator further provides skillful seasonal temperature forecasts at 1–12 month lead times and permits analysis of changes in wind-speed extremes (10-year return levels) under modified forcing. Beyond projections, GCMagicc enables the study of emergent constraints and teleconnections across the multi-model ensemble. We present case studies on global breadbasket analysis and global drought trends to illustrate some potential applications.

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